

§ 11.4 The cross product.

$$\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad \vec{u} \times \vec{v} = \langle u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1 \rangle$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Recall that the 2×2 determinant is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad \text{eg} \quad \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = 1 - 6 = -5$$

and 3×3 determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ is given by

$$a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

$$\text{so } \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

$$= \vec{i}(u_2 v_3 - u_3 v_2) - \vec{j}(u_1 v_3 - u_3 v_1) + \vec{k}(u_1 v_2 - u_2 v_1)$$

$$= \langle u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1 \rangle$$

$$(1, 2, 0) \times (2, -1, 1) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 0 \\ 2 & -1 & 1 \end{vmatrix} = (2 - 0, -1 + 0, -1 - 4) = (2, -1, -5)$$

$$(2, -1, 1) \times (1, 2, 0) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix} = (0 - 2, 0 + 1, 4 + 1) = (-2, 1, 5)$$

$$\begin{vmatrix} \textcircled{a_1} & \cancel{a_2} & \cancel{a_3} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\begin{vmatrix} \cancel{a_1} & \textcircled{a_2} & \cancel{a_3} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\textcircled{1} \quad \vec{u} \times \vec{v} = -(\vec{v} \times \vec{u}) \quad \text{anti-commutative}$$

$$\textcircled{2} \quad \vec{u} \cdot (\vec{u} \times \vec{v}) = 0 = \vec{v} \cdot (\vec{u} \times \vec{v}) \quad \text{perpendicular to } \vec{u}, \vec{v}$$

$$\textcircled{3} \quad \vec{u}, \vec{v}, \vec{u} \times \vec{v} \text{ are a right handed triple}$$

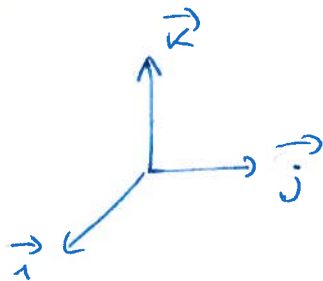
$$\textcircled{4} \quad \|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \theta$$

pt $\textcircled{2} \quad \vec{u} \cdot (\vec{u} \times \vec{v}) = (u_1, u_2, u_3) (u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1)$
 $= \underbrace{u_1 u_2 v_3} - \underbrace{u_1 u_3 v_2} + \underbrace{u_2 u_3 v_1} - \underbrace{u_2 u_1 v_3} + \underbrace{u_3 u_1 v_2} - \underbrace{u_3 u_2 v_1} = 0$

$$\textcircled{3} \quad \text{eg} \quad \vec{i} \times \vec{j} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \vec{i} \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = \vec{k}$$

$$\vec{j} \times \vec{k} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \vec{i} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = \vec{i}$$

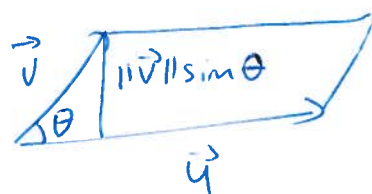
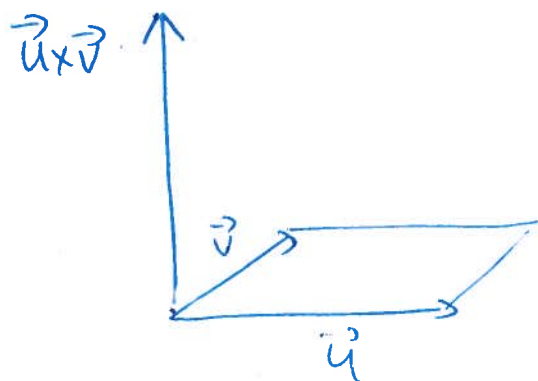
$$\vec{k} \times \vec{i} = \vec{j}$$



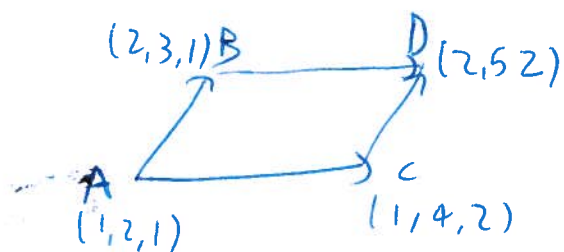
$$\textcircled{4} \quad \text{eg} \quad \|\vec{u} \times \vec{u}\| = \|\vec{u}\|^2 \sin 0 = 0 \quad \text{so } \vec{u} \times \vec{u} = \vec{0}$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ u_1 & u_2 & u_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ u_2 & u_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ u_1 & u_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ u_1 & u_2 \end{vmatrix} \\ = \vec{i} (u_2 u_3 - u_2 u_3) - \vec{j} (u_1 u_3 - u_1 u_3) + \vec{k} (u_1 u_2 - u_1 u_2) \\ = \vec{0}$$

Ex $\vec{u} \times \vec{v}$ = vector perp to plane spanned by $\vec{u}, \vec{v}$ of length equal to area of parallelogram



Find area of parallelogram $(1, 2, 1), (2, 3, 1)$
 $(1, 4, 2), (2, 5, 2)$



$$\vec{AB} = (1, 1, 0) \quad \vec{BD} = (0, 2, 1)$$

$$\vec{AC} = (0, 2, 1) \quad \vec{CB} = (1, 1, 0)$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{vmatrix} =$$

$$= \vec{i} \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = (1, -1, 2)$$

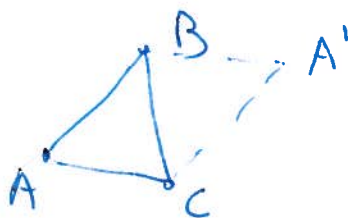
$$\text{Area} = \|(1, -1, 2)\| = \sqrt{1+1+4} = \sqrt{6}$$

$$\sin \theta = \frac{\sqrt{6}}{\sqrt{2} \cdot \sqrt{5}} = \sqrt{\frac{3}{5}}$$

$$\theta = \arcsin \sqrt{\frac{3}{5}}$$

Ex Find area of ΔABC where

$$A = (1, 2, 1) \quad B = (3, 3, 3) \quad C = (2, 4, 4)$$

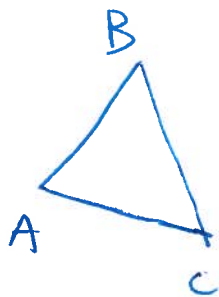


$$\vec{AB} = (2, 1, 2)$$

$$\begin{aligned} \text{Ex } \vec{u} \cdot (\vec{v} \times \vec{w}) &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (u_1(v_2w_3 - v_3w_2), -u_2(v_1w_3 - v_3w_1), u_3(v_1w_2 - v_2w_1)) \\ &= \left(u_1^2(v_2w_3 - v_3w_2)^2 + u_2^2(v_1w_3 - v_3w_1)^2 + u_3^2(v_1w_2 - v_2w_1)^2 \right)^{1/2} \end{aligned}$$

Eg Find area of $\triangle ABC$ where

$$A = (1, 2, 1) \quad B = (3, 3, 3) \quad C = (2, 4, 4)$$

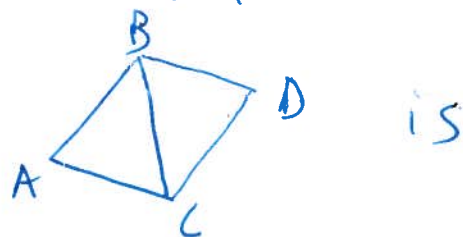


$$\vec{AB} = (2, 1, 2)$$

$$\vec{AC} = (1, 2, 3)$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = (3-4, -6+2, 4-1) = (-1, -4, 3)$$

Area of parallelogram

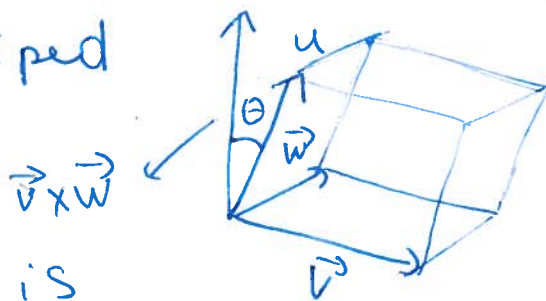


$$\sqrt{1+16+9} = \sqrt{26}$$

Area of triangle is $\frac{1}{2} \times \sqrt{26}$

$$\underline{\text{Eg}} \quad \vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = u_1(v_2w_3 - v_3w_2) - u_2(v_1w_3 - v_3w_1) + u_3(v_1w_2 - v_2w_1)$$

This is the volume (with sign) of the parallelepiped



Area of the base is

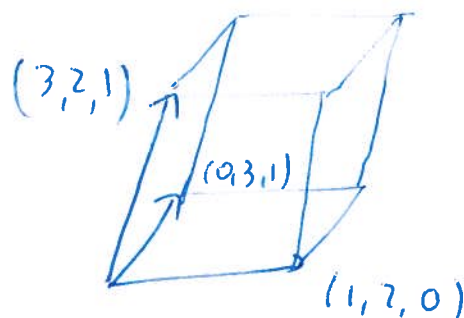
$\|\vec{v} \times \vec{w}\|$, height is $\|u\| \cos \theta$

$$\text{Volume is } \pm \|\vec{u}\| \|\vec{v} \times \vec{w}\| \cos \theta = \pm \vec{u} \cdot (\vec{v} \times \vec{w})$$

Ex $\vec{i} \cdot (\vec{j} \times \vec{k}) = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$

Ex $(3, 2, 1) \cdot ((1, 2, 0) \times (0, 3, 1)) = (3, 2, 1) \cdot \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 0 \\ 0 & 3 & 1 \end{vmatrix} =$
 $= (3, 2, 1) \cdot (2, -1, 3) = 6 - 2 + 3 = 7$

This is the volume of
the parallelepiped



Ex Compute $(1, t, -t) \times (t, t, -2)$

Ex Find the unit vectors perpendicular to
the plane determined by $(1, 3, 5), (3, -1, 2), (4, 0, 1)$.

Ex Find the area of the triangle with vertices
 $(3, 2, 1), (2, 4, 6), (-1, 2, 5)$

Ex Find the eqn of the plane through $(7, 0, 0), (0, 3, 0)$
and $(0, 0, 5)$

Ex Find the eqn of the plane through $(2, -3, 2)$
and parallel to $(4, 3, -1)$ & $(2, -5, 6)$

Ex Find the volume of the parallelepiped
with edges $(2, 3, 4), (0, 4, -1)$ & $(5, 1, 3)$